Exploring EEG classification and EEG-Based Caption Retrieval

Presented by

Ahmed S Abouhashem
Nayesha Gandotra

When humans view images, their brains generate distinctive patterns of electrical activity that can be measured using electroencephalography (EEG)

Can we decode what someone is seeing from their brain activity alone?

Outline

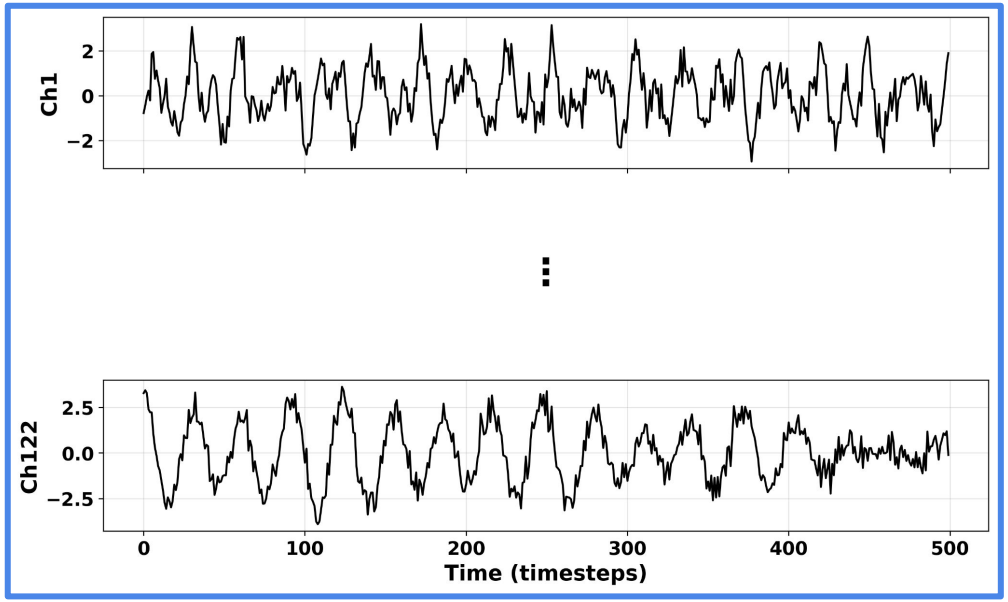
Overview

EEG encoder for classification

Caption retrieval

- Image-Caption
- EEG-Caption

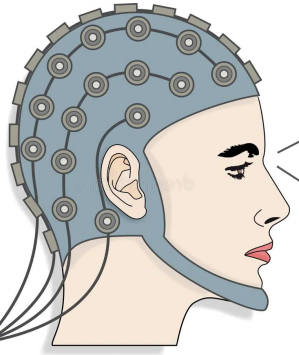
Overview: EEG

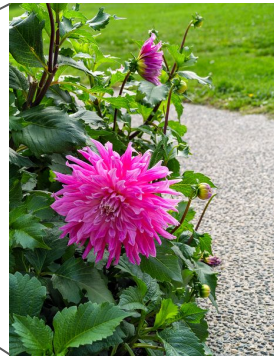


EEG records brain activity through electrodes

measures voltage fluctuations reflecting the synchronous firing of cortical neurons

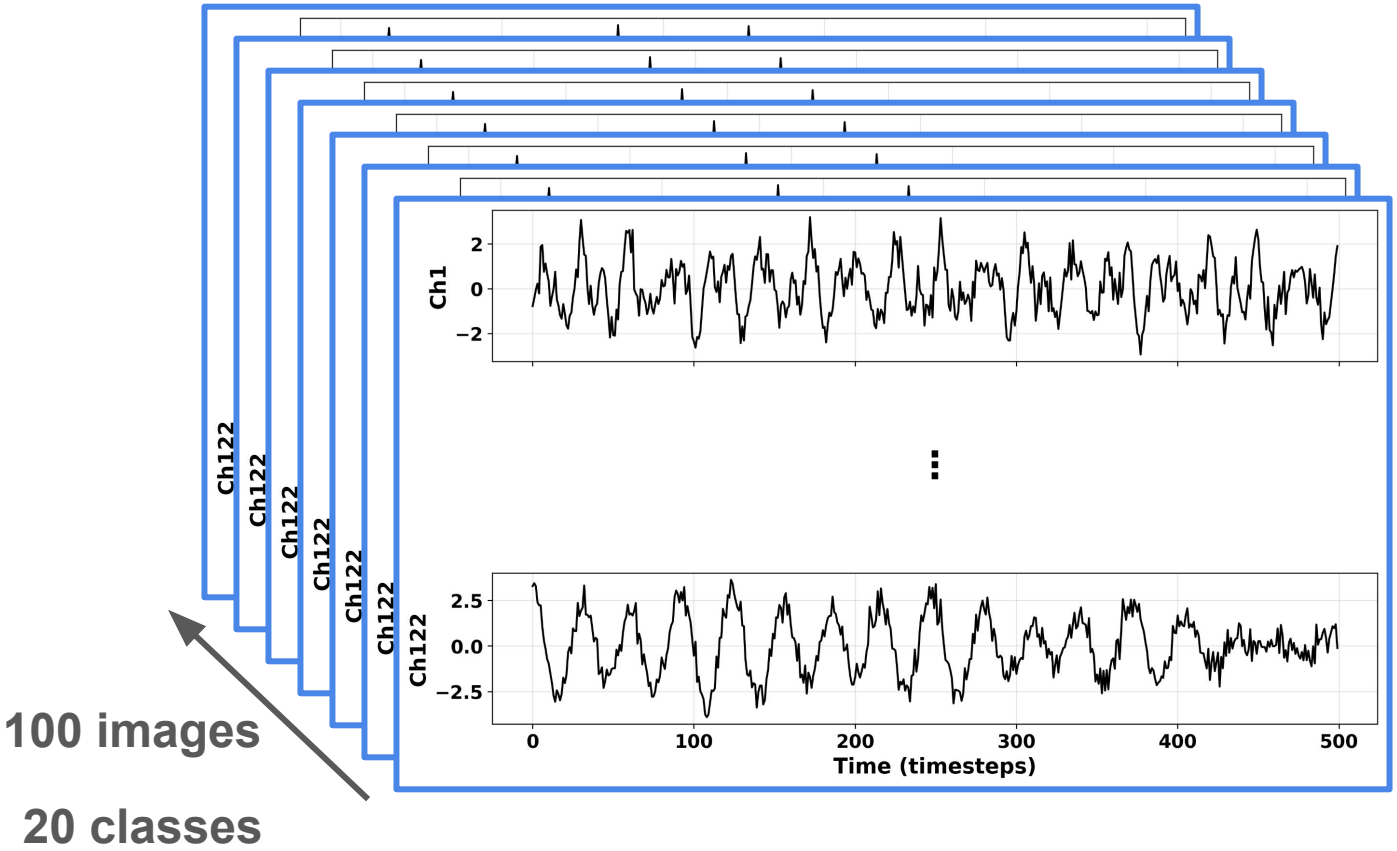
Bandpass filtering:
0.1 Hz: removes very slow drifts
100 Hz: removes very fast fluctuations (e.g., muscle activity)



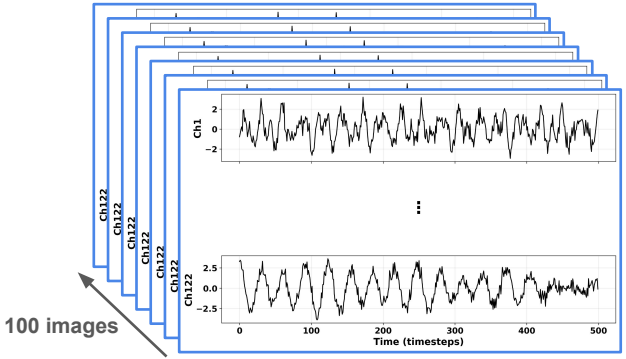


EEG: Electroencephalogram

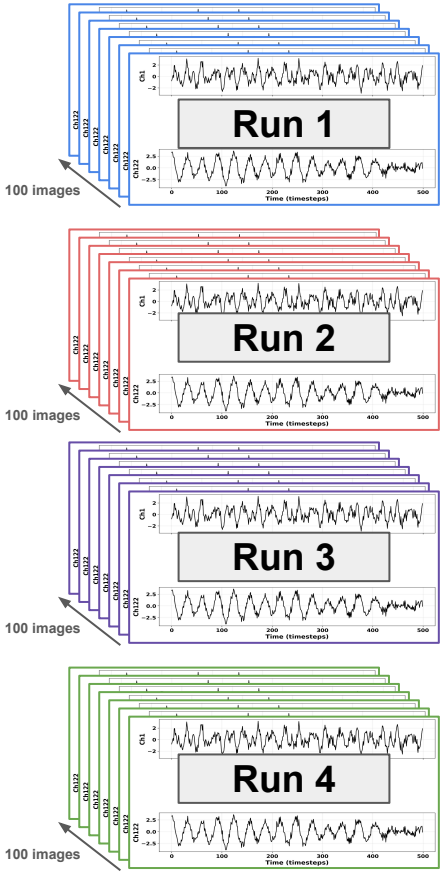
Each run (low-speed paradigm)



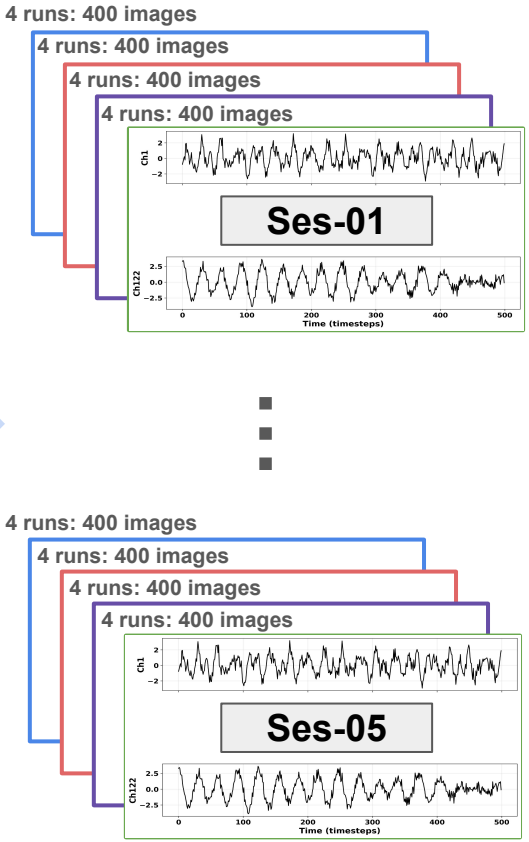
Data from 1 Run



Data from 1 Session



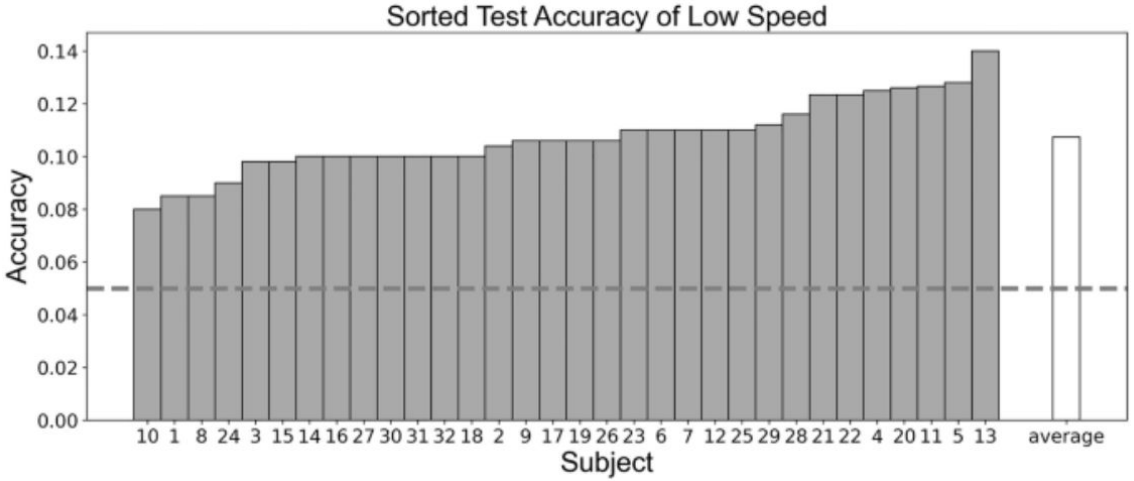
Data from 1 Subject

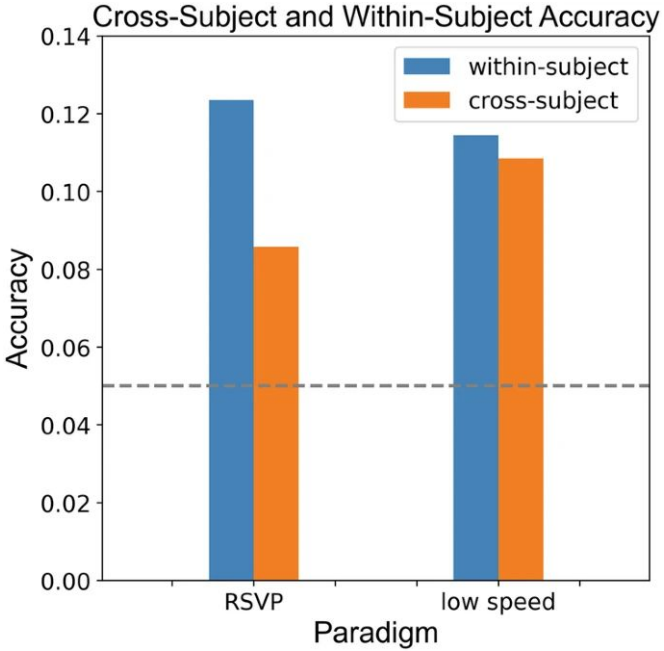


EEG-based Image Classification

From original paper:

Within-session accuracy: Train on run 1, 2 and 3 & Test on run 4





https://www.nature.com/articles/s41597-025-04843-x#Fig3 (Xue et al. 2025)

Their model

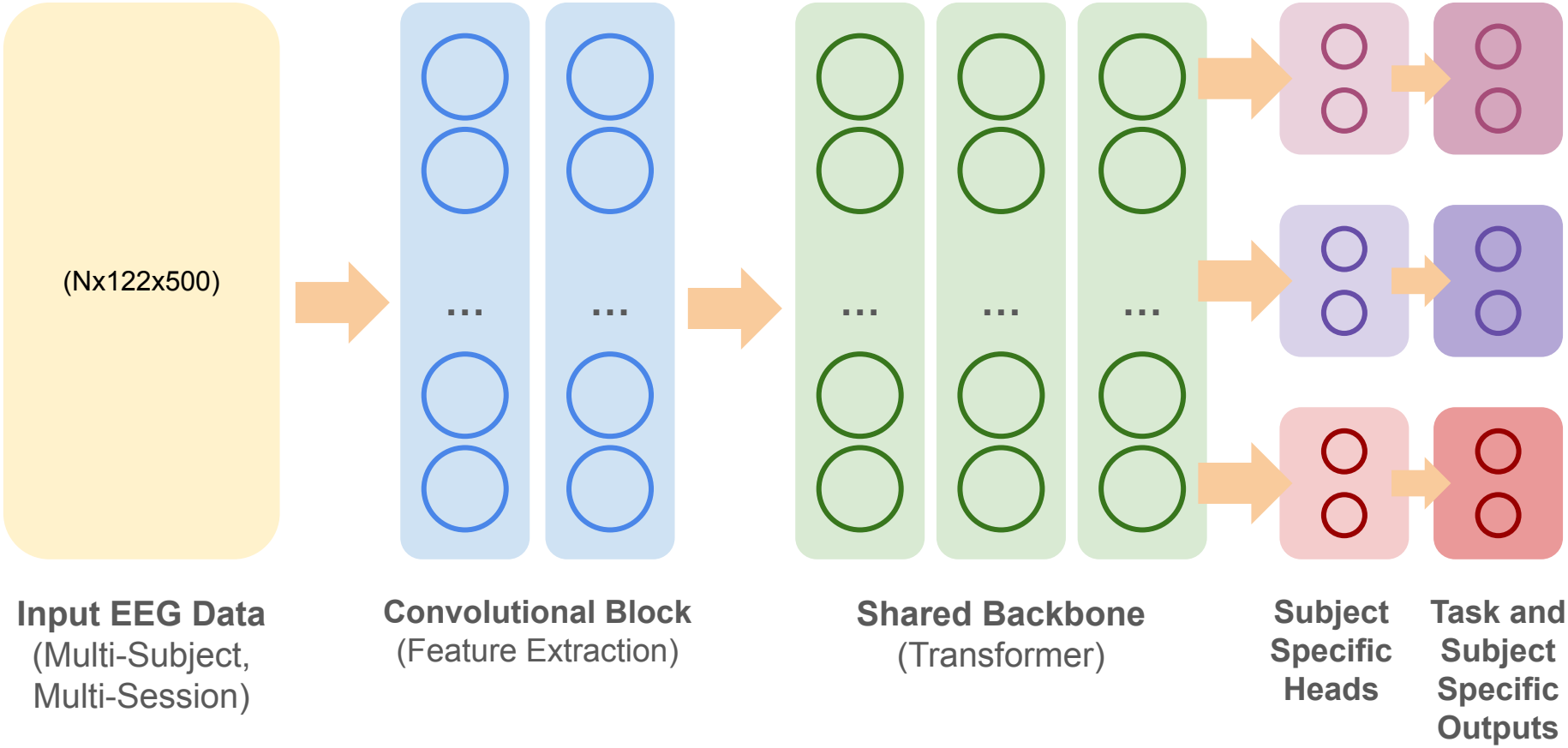
- model_conv.py
- model_fc.py
- model_linear.py
- model_lstm.py
- model_transformer.py

```python
from torch import nn

class ModelFC(nn.Module):
    def __init__(self, in_dim, hid_dim, out_dim,):
        super(ModelFC, self).__init__()
        self.model = nn.Sequential(
            nn.Linear(in_dim, hid_dim),
            nn.BatchNorm1d(num_features=hid_dim),
            nn.ReLU(),
            nn.Linear(hid_dim, out_dim),
            nn.Softmax(dim=-1),
        )
    def forward(self, x):
        # bs * time * electrode
        x_flat = x.reshape(x.shape[0], -1)
        return self.model(x_flat)
```

https://github.com/xuesn/EEGDataset (Xue et al. 2025)

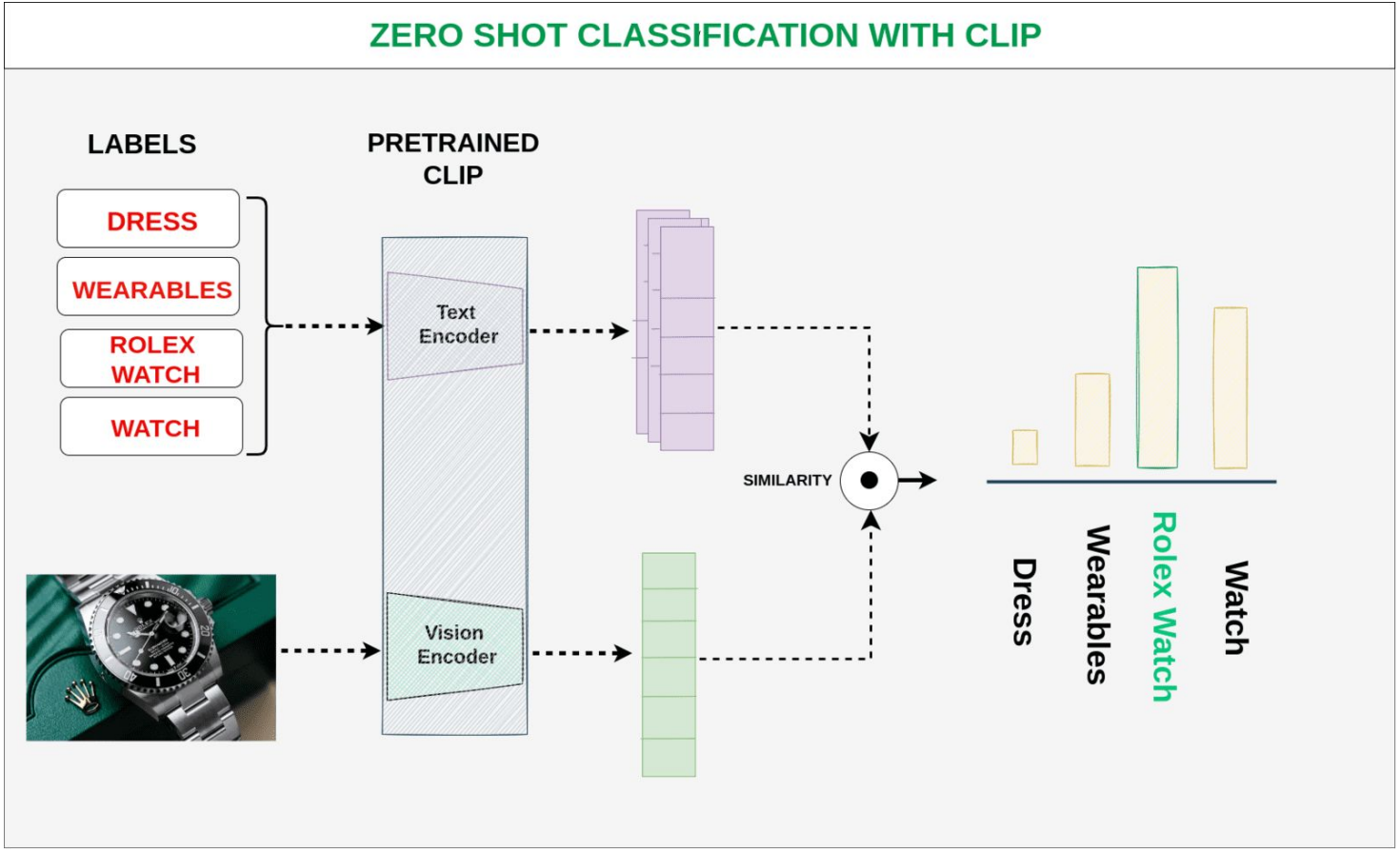
The model you will work on



Caption Retrieval

CLIP (Contrastive Language–Image Pretraining)

https://arxiv.org/abs/2103.00020 (Radford et al.)



https://learnopencv.com/clip-model/

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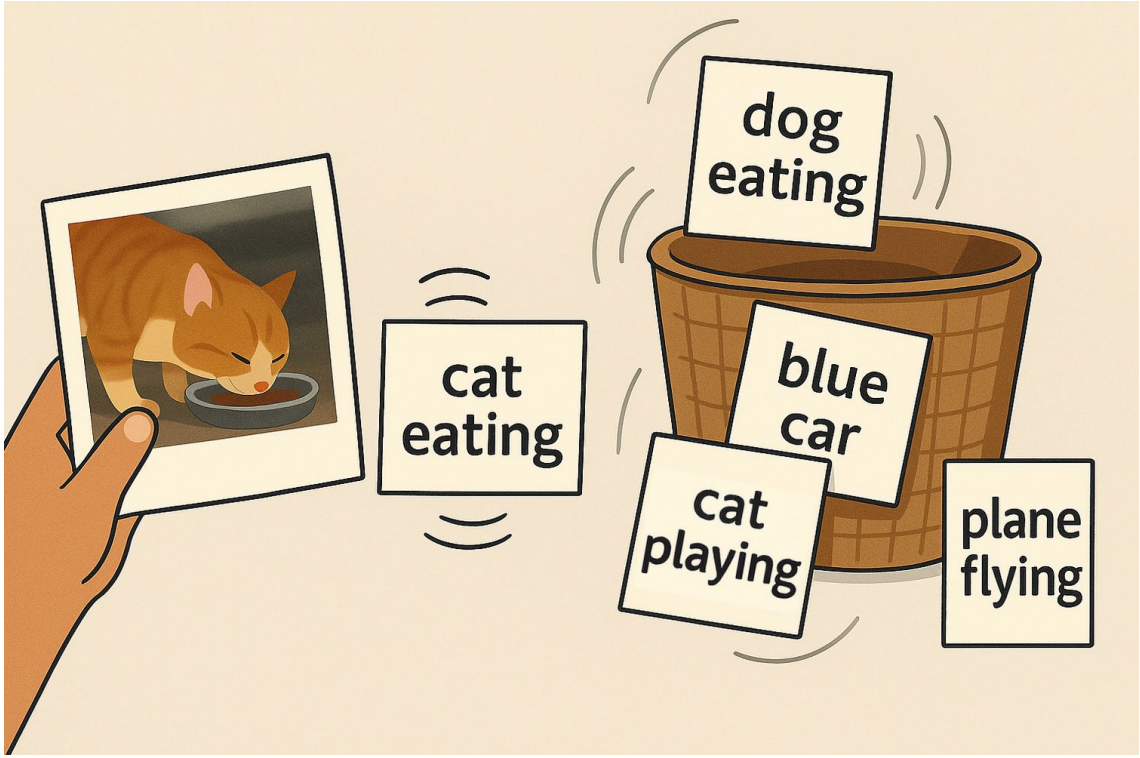


Different applications:

- Image-caption retrieval
- Zero-shot classification
- Cross-domain tasks such as in robotics (video-text)
- Medical applications (pathology and radiology)

To-Do: Use CLIP for Image-Caption retrieval

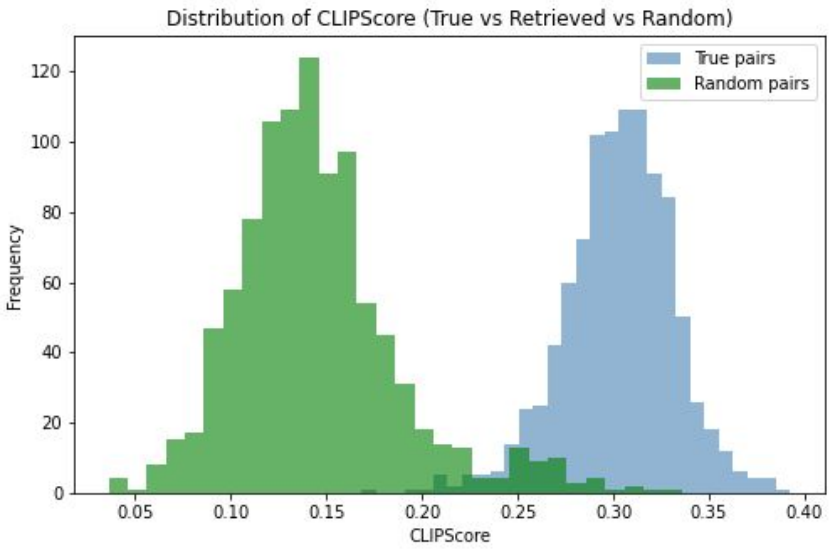
Download a pretrained CLIP model (frozen model - no training)

Image and caption embedding

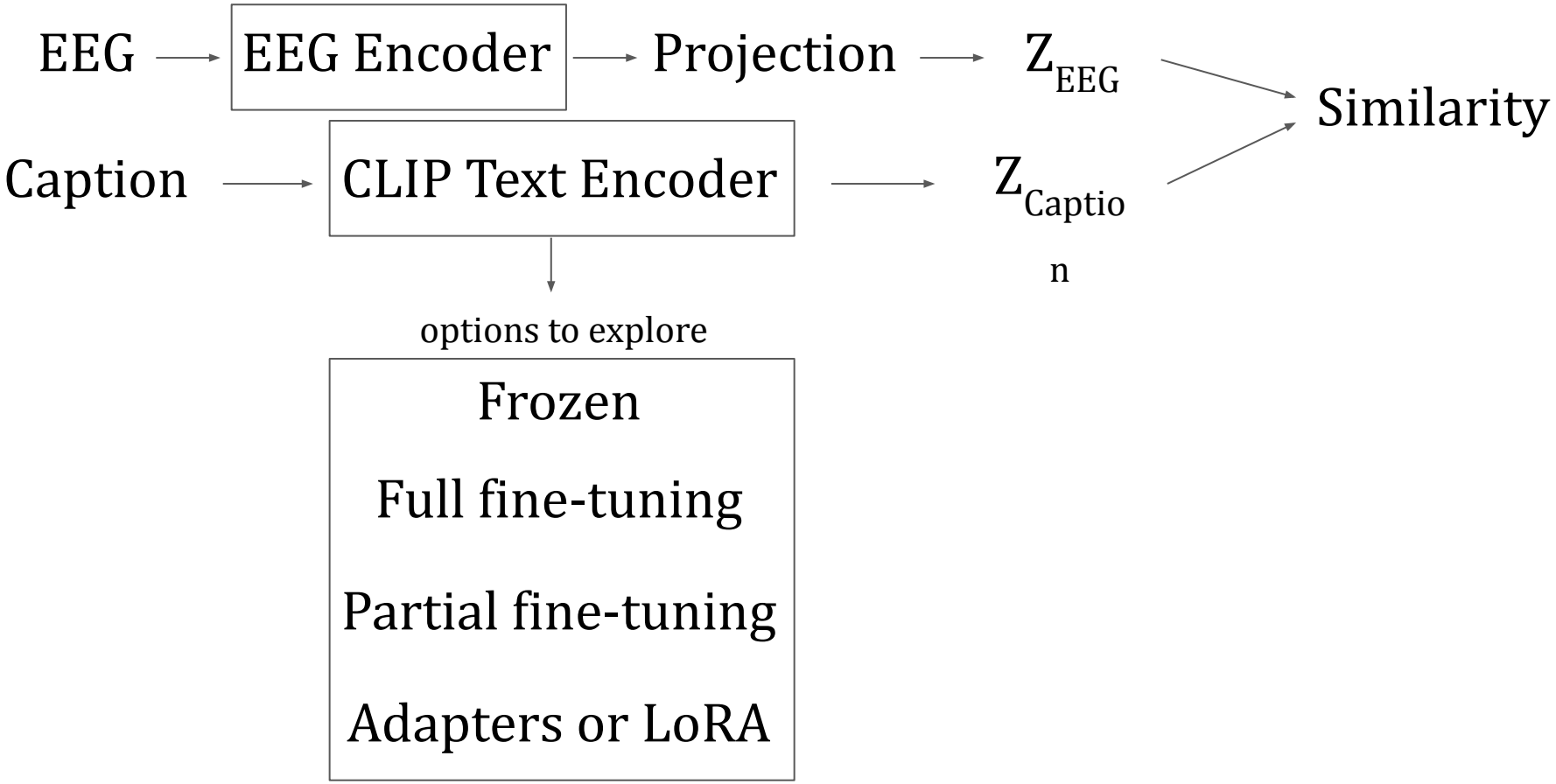
Use image embeddings to retrieve the caption with highest similarity

Evaluate the retrieved caption:

- BERTScore
- Recall@1, 3 and 5
- Class-aware Recall@1,3 and 5
- Mean Average Precision
- Similarities of random mismatches



To-Do: EEG-Caption retrieval



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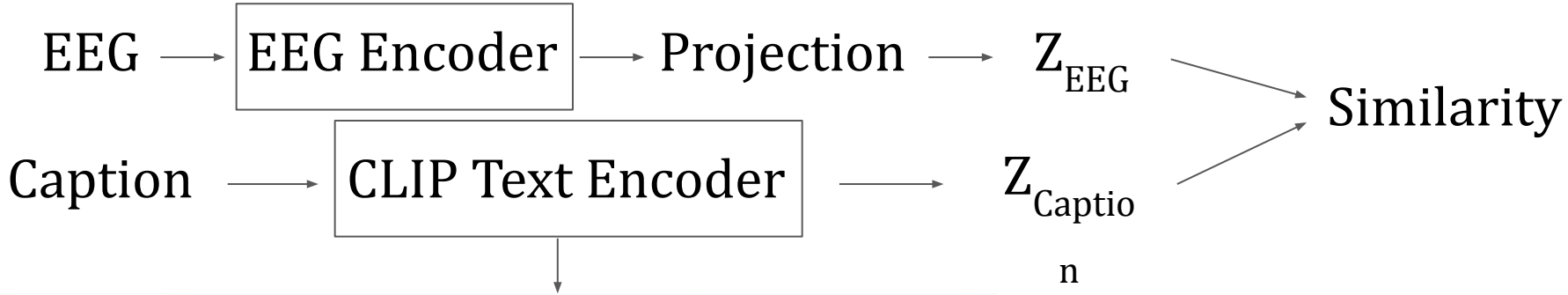
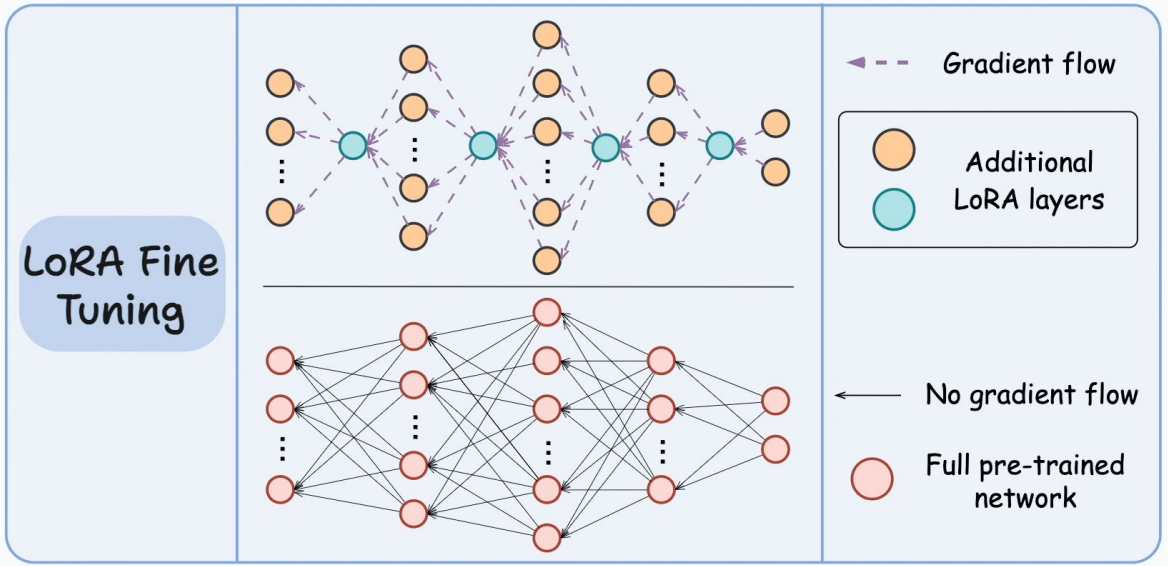
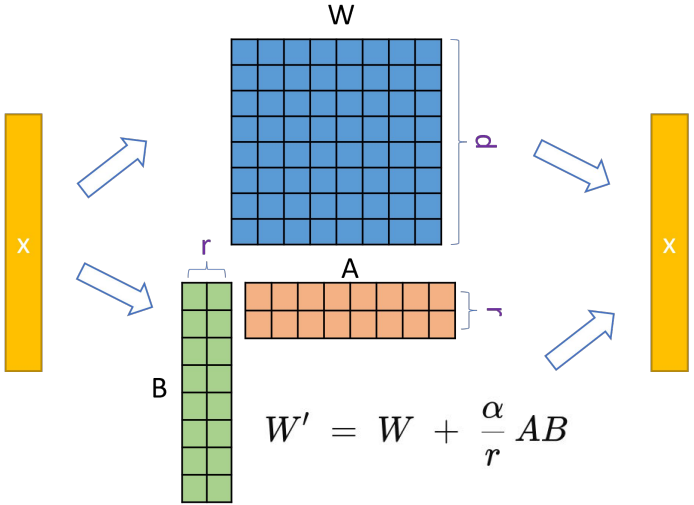




https://blog.dailydoseofds.com/p/full-model-fine-tuning-vs-lora-vs



https://tursunboev.com/posts/2023-05-30-low-rank-adaptation-of-large-llms/

To-Do: EEG-Caption retrieval

Loss function Knowledge Distillation

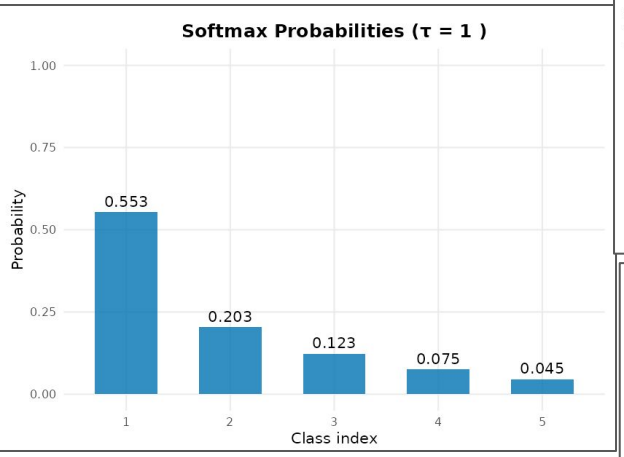
$$\mathcal{L}_{\cos} = 1 - \mathbf{eeg}_z^\top \mathbf{txt}_z$$

$$q_j = \frac{\exp(\mathrm{sim}(\mathbf{img}_z, \mathbf{txt}_{z,j})/\tau_t)}{\sum_{k=1}^{M} \exp(\mathrm{sim}(\mathbf{img}_z, \mathbf{txt}_{z,k})/\tau_t)}, \quad p_j = \frac{\exp(\mathrm{sim}(\mathbf{eeg}_z, \mathbf{txt}_{z,j})/\tau_s)}{\sum_{k=1}^{M} \exp(\mathrm{sim}(\mathbf{eeg}_z, \mathbf{txt}_{z,k})/\tau_s)}.$$

$$\mathrm{KL}(q \,\|\, p) = \sum_{j=1}^{M} q_j \log \frac{q_j}{p_j}.$$

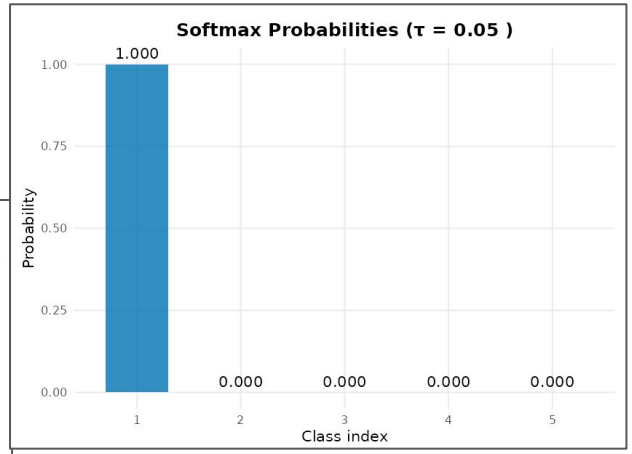
Effect of τ

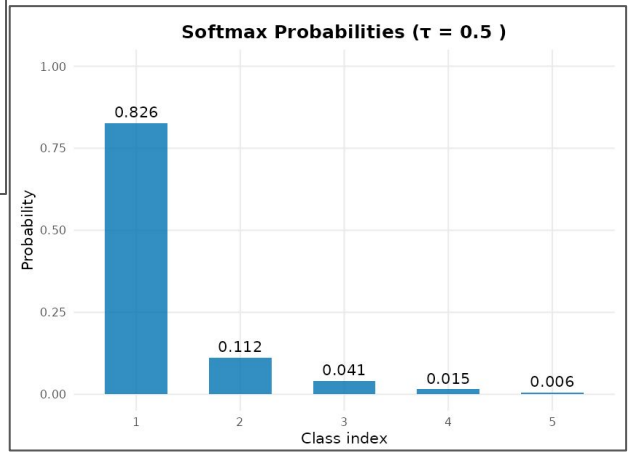
$T = 1$



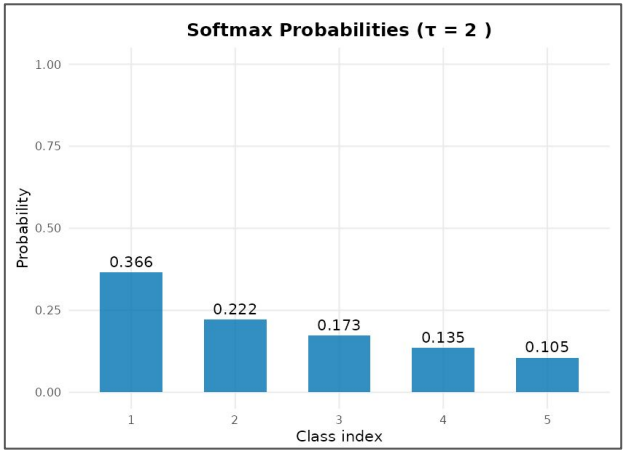
$$p_i = \frac{\exp\left(\frac{z_i}{\tau}\right)}{\sum_{j=1}^{N} \exp\left(\frac{z_j}{\tau}\right)}$$

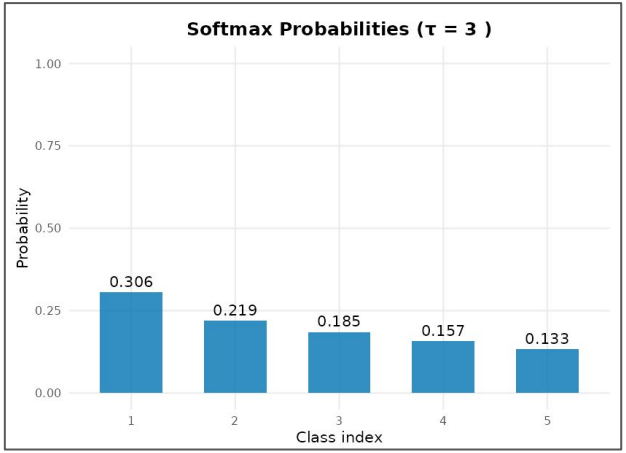
low τ





high τ





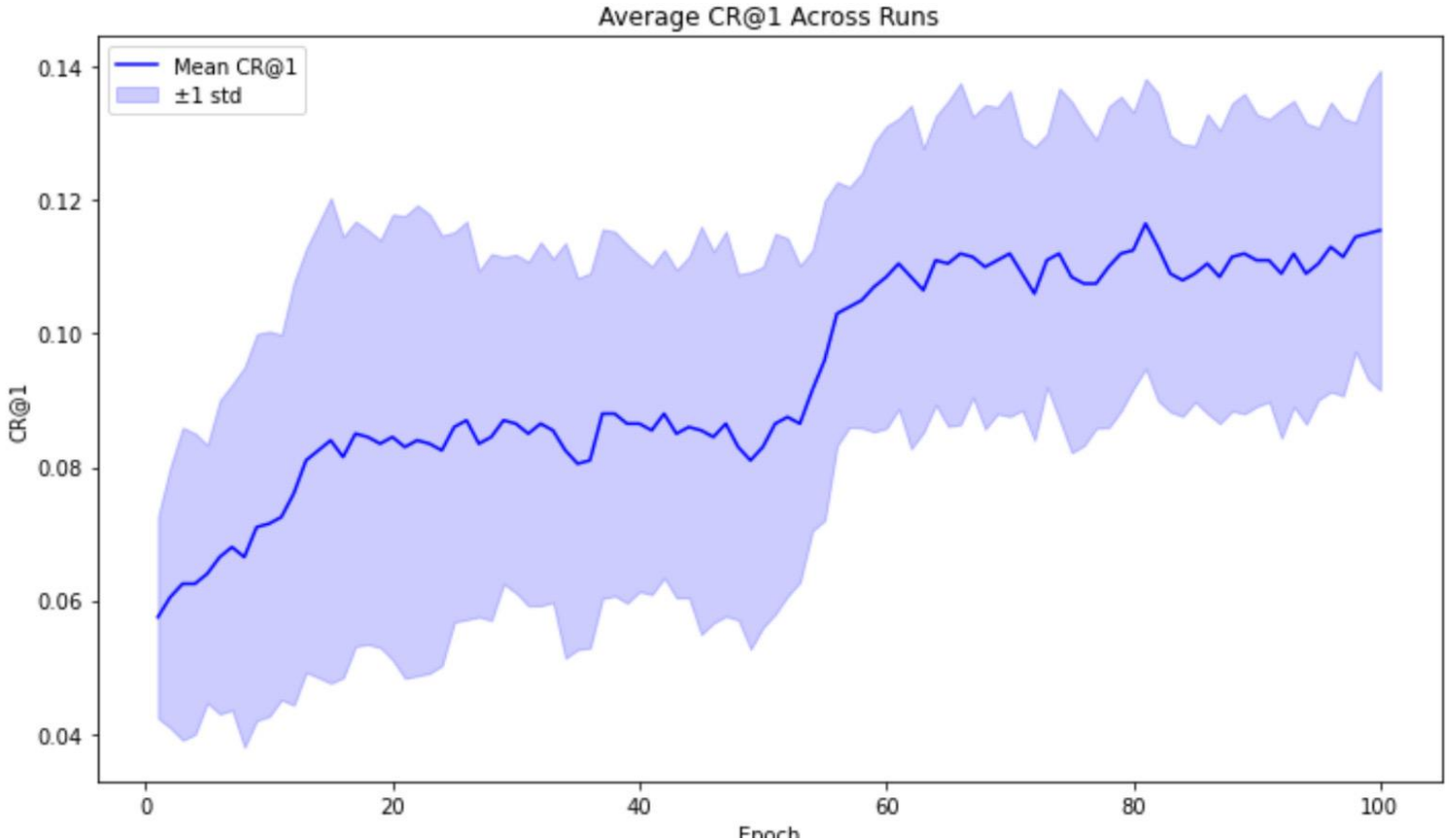
To-Do: EEG-Caption retrieval

Loss function Contrastive Alignment

$$\mathcal{L}_{\mathrm{NCE}}^{\mathrm{E}\to\mathrm{T}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(Z_{ii})}{\sum_{j=1}^{N}\exp(Z_{ij})}.$$

$$\mathcal{L}_{\mathrm{debias}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(Z_{ii})}{\sum_{j=1}^{N}\boxed{w_{ij}}\exp(Z_{ij})}.$$

Preliminary results using baseline (class-level)



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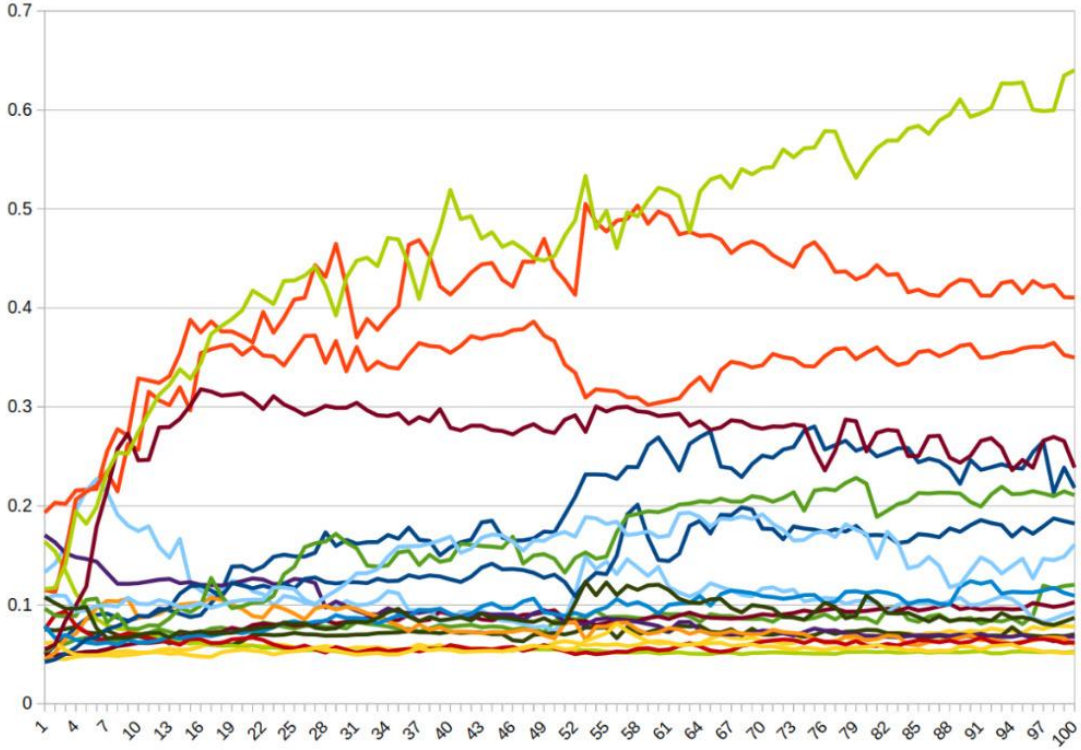

Top classes by MAP (last 5 epochs)

MAP_class_tvmonitor	0.614731
MAP_class_bicycle	0.418578
MAP_class_horse	0.357711
MAP_class_sheep	0.255629
MAP_class_aeroplane	0.237963
MAP_class_person	0.212577
MAP_class_flower	0.181067
MAP_class_sofa	0.145644
MAP_class_dog	0.113147
MAP_class_boat	0.112840
MAP_class_bottle	0.100417
MAP_class_bus	0.087628
MAP_class_train	0.083256
MAP_class_bird	0.077802
MAP_class_car	0.068824
MAP_class_chair	0.066203
MAP_class_cow	0.064814
MAP_class_diningtable	0.063731
MAP_class_motorbike	0.052536
MAP_class_cat	0.052417

Tasks

1. Task 1: Classification
    a. Baseline model (from the original paper, EEGNet or any other model)
    b. Implement a model (backbone + heads)
    c. Setup: Train on 3 sessions, validate on 1 and test on 1.
    d. Evaluate classification: accuracy and confusion matrix

1. Task 2: Caption Retrieval
    b. CLIP for Image-Caption retrieval: Frozen text and image encoder
    c. Evaluate Image-retrieved captions:
        i. Semantic metrics
        ii. Recall and precision
        iii. CLIPScore similarities between paired matches and mismatches
    d. CLIP for EEG-Caption retrieval: Frozen, fine-tuning (partial unfreezing, LoRA, … etc)
    e. Evaluate EEG-retrieved captions:
        i. Recall and precision
        ii. CLIPScore similarities between paired matches and mismatches

Thanks!